

Surrogate Substitution Preserves PHI Detectability: A Multi-Detector Equivalence Study

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Abstract

Structure-preserving de-identification replaces protected health information (PHI) with realistic same-type *surrogates*—“Anna S.” becomes “Maria S.”, not [NAME]—so that clinical text stays fluent and downstream tools keep working. But this only helps if the substitution does not itself corrupt the signal those tools rely on. We ask a narrow, testable question: *on the spans a de-identifier actually masks, can downstream PHI detectors still find the surrogate?* We introduce a paired, multi-detector evaluation protocol that (i) scores utility *only on masked spans*, decoupling *coverage* from *utility*; (ii) uses *equivalence testing* (TOST) rather than null-hypothesis significance testing, which is uninformative at our sample size (57k paired spans); and (iii) builds a surrogate-failure typology separating fixable generator defects from intrinsic detector limits. Across 11 detectors, 7 benchmarks, and 7 languages (1,750 documents), recall on masked spans moves from 76.1% to 74.9%—a change our equivalence test shows is *statistically equivalent to zero within a ± 2 -point margin* ($p \approx 3 \times 10^{-9}$), with detector ranking preserved. The residual loss does not reflect detectors getting worse at PHI: it concentrates in *malformed and out-of-distribution* surrogates (truncation *Chicago* $\rightarrow$ *Illino*, salience loss *Cedars-Sinai* $\rightarrow$ *Vidant*). A redaction floor and an open-source surrogate baseline indicate the effect is a property of well-formed substitution, not of one tool. We release the evaluation subsets, scoring code, and an interactive dashboard (URLs withheld for anonymous review) so the protocol can audit any structure-preserving transform.

1. Introduction

De-identification of clinical text takes two forms—and only one keeps the text usable downstream. **Redaction** deletes or tags PHI (*John Smith* $\rightarrow$ [NAME]), which is safe but destroys the fluency, layout, and distributional properties that downstream models and human readers depend on. **Structure-preserving** de-identification instead substitutes each PHI value with a plausible, same-type *surrogate* (*John Smith* $\rightarrow$ *Maria Lopez*), keeping the document readable and machine-parseable (Carrell et al., 2013). The second family is increasingly attractive: it lets de-identified data flow into analytics, model training, and even second-pass detection without breaking pipelines built for real text.

The promise of structure preservation rests on an unstated assumption: *the surrogate carries the same downstream signal as the value it replaced*. If substitution silently degrades the very features a PHI detector, an NER model, or a clinical parser relies on, then “structure-preserving” is a misnomer—the transform would be quietly laundering PHI into forms tools can no longer see or handle, which is both a utility problem and, for a second-pass safety net, a privacy problem.

This assumption is rarely tested directly, and testing it well is harder than it looks. Three pitfalls recur:

1. **Coverage confounds utility.** A de-identifier that masks 60% of PHI and one that masks 95% cannot be compared on whole-document F1—the score conflates *how much* it masks with *whether what it masks stays usable*. The two must be measured separately.
2. **The large- N significance trap.** With tens of thousands of paired spans, any non-zero difference is “statistically significant” under a standard test, even when operationally mean-

ingless. A p -value here answers the wrong question.

3. **Aggregate error hides mechanism.** A single “−1.2 points” says nothing about *why* the loss happens—boundary jitter, detector weakness, or a defect in surrogate generation. Only the last is fixable, and only an error typology tells them apart.

We address all three. Our contributions:

- A **paired multi-detector protocol** scoring utility on masked spans only, decoupling coverage from utility, across 11 detectors $\times$ 7 benchmarks $\times$ 7 languages (§4–§5).
- An **equivalence-testing analysis** (TOST) replacing the uninformative significance test with a bounded-effect claim: the change in detectability is bounded within ± 2 points of zero (§6).
- A **surrogate-failure typology** attributing the small residual to surrogate-generation quality rather than detectors getting worse at PHI (§7).
- A set of **comparison experiments**—a redaction upper-bound and an open-source surrogate baseline—that make the result reproducible and isolate what is generic to *any* substitution versus specific to one tool (§5, §6).

The framing is deliberately not “a proprietary tool is good.” It is: *here is how to measure whether a structure-preserving transform preserves utility, rigorously*, with one commercial transform as the case study and open baselines for reproducibility.

2. Related Work

2.1. De-identification and surrogate substitution

Clinical de-identification is a long-studied sequence-labeling problem (Uzuner et al., 2007), anchored by the i2b2/n2c2 and MEDDOCAN shared tasks (Stubbs and Uzuner, 2015; Marimon et al., 2019) and by systems ranging from rule- and dictionary-based pipelines (Microsoft, 2018) through recurrent and transformer taggers (Deroncourt et al., 2017; Liu et al., 2017; Johnson

et al., 2020) to instruction-tuned LLMs used zero-shot (Liu et al., 2023). Most of this literature optimizes *detection*, typically against the HIPAA Safe Harbor identifier set (U.S. Department of Health and Human Services, 2012). The downstream question—*what to put in the PHI’s place*—is comparatively under-studied. Surrogate (“hiding in plain sight”) replacement was proposed to keep de-identified notes realistic and to resist re-identification (Carrell et al., 2013); subsequent work showed poor surrogates can even aid re-identification (the “parrot” attack), underscoring that surrogate *quality* matters (Carrell et al., 2020). Our work is orthogonal to the detection literature and to any particular surrogate generator: given that a span is masked, we ask whether the *replacement* remains detectable and well-formed.

The commercial transform used as our case study is a proprietary, structure-preserving de-identification API (vendor and product name withheld for anonymous review); we treat it as the source of the transform rather than describing or extending the method.

2.2. Utility-preservation evaluation

Whether privacy transformations preserve downstream utility is central to privacy-preserving NLP. Prior evaluations typically run a single downstream model on original vs. transformed data and compare end-task scores. Two weaknesses recur: (a) a single downstream model cannot separate “the transform is fine” from “this model is robust,” and (b) whole-corpus metrics mix masked and unmasked content. We address (a) with an 11-detector panel spanning rule-based, fine-tuned, and LLM detectors, and (b) by restricting utility measurement to masked spans.

2.3. The large- N trap and equivalence testing

When samples are large, null-hypothesis significance testing rejects the null for negligible effects; the p -value measures precision, not importance—a hazard increasingly flagged in NLP evaluation (Dror et al., 2018; Card et al., 2020). The standard remedy is *equivalence testing*—the two one-sided tests (TOST) procedure (Schuirmann, 1987; Berger and Hsu, 1996)—which specifies an equivalence margin Δ and tests whether the effect lies inside $[-\Delta, +\Delta]$; see Lakens (2017); Lakens

et al. (2018) for practical tutorials. TOST is standard in biostatistics but under-used in NLP evaluation, where large paired corpora make the trap acute. We adopt it as the primary inferential tool and report McNemar’s paired test (McNemar, 1947) only to demonstrate the trap.

3. Problem Formulation

Structure-preserving de-identification. A transform T maps a document d to d' by replacing each detected PHI value v (of type τ) with a surrogate s of the same type, leaving all other characters unchanged. Gold PHI spans on d are re-projected onto d' by character-level alignment, giving paired spans (v, s) .

Coverage vs. utility. Two quantities must not be conflated. *Coverage* is the fraction of true PHI that T detects and replaces (a property of T ’s detector). *Utility* is, given that a span was replaced, whether a downstream detector still finds the surrogate s as well as it found v (a property of T ’s generator and of the surrogate’s realism). We report coverage separately and measure utility *only on the masked-span population* $\{(v, s)\}$. This prevents a low-coverage transform from looking good—or a high-coverage one from looking bad—on a metric that is really about substitution quality.

4. Evaluation Protocol

Figure 1 summarizes the protocol. **Paired multi-detector design.** Each document is scored in two conditions—original d and transformed d' —by the *identical* detector suite D ($|D| = 11$). Because the only change between conditions is the substitution, any per-span change in detection is attributable to the substitution, not to the detector or the document.

Metrics. We report span-level P/R/F1 under three matching modes—*exact* (start+end+type), *type* (type+boundary), *overlap* (any character overlap+type)—and *leakage* = $1 - \text{recall}$, the HIPAA-critical quantity. The headline utility metric is *recall retention on masked spans* = $\text{recall}(d') / \text{recall}(d)$ restricted to $\{(v, s)\}$, under overlap matching (so pure boundary jitter from length changes, “Anna S.” → “Maria S.”, is not charged as a miss).

Equivalence testing. For pooled masked-span recall we run TOST with margin $\Delta = 2$ points: we reject non-equivalence iff the 90% CI of the recall difference lies entirely within $[-2, +2]$. We also report McNemar’s test to illustrate the large- N trap. We sweep $\Delta \in \{1, 2, 3\}$.

Error attribution. For the lost population (found on d , missed on d') we hand-code each span into a small failure typology (§7) and check length-preservation to separate boundary artifacts from genuine misses.

5. Experimental Design

Detector panel D . Eleven detectors, three families: *rule/statistical*—Microsoft Presidio (Microsoft, 2018) (built on spaCy (Honnibal et al., 2020)), OBI deid_roberta (Johnson et al., 2020), a RoBERTa tagger (Liu et al., 2019; Devlin et al., 2019); *open LLMs (local)*—Gemma 4 31B / E4B (Gemma Team, Google DeepMind, 2024), Qwen 3.5-{4B, 9B, 35B-A3B} (Qwen Team, 2024), Llama 3.1-8B / 3.3-70B (Grattafiori, Aaron and Dubey, Abhimanyu and others (Llama Team, AI @ Meta), 2024), DeepSeek V2-Lite (DeepSeek-AI, 2024); *frontier API*—OpenAI GPT-5 (OpenAI, 2023). All are Transformer-based (Vaswani et al., 2017). The panel is deliberately heterogeneous: if the equivalence result held only for one architecture it would be a model artifact, not a property of the transform. (Model-version citations point to the closest published technical report for each family. One further model, Moonshot Kimi-VL-A3B, was attempted but excluded because it required non-standard model-loading code.) LLM detectors emit free-text or JSON identifier lists; we recover character spans by exact-then-fuzzy string search over the source document, so all detectors are scored on the same span basis.

Benchmarks (7; 250 docs each, 1,750 total). ASQ-PHI (English clinical queries) (Weatherhead et al., 2026), MEDDOCAN (Spanish clinical) (Marimon et al., 2019), MultiCoNER v2 (multilingual NER) (Malmasi et al., 2022; Fetahu et al., 2023), and PII-Masking-300k (ai4Privacy, 2023) in English, Dutch, French, German. Together: 7 languages, clinical and general-PII text, free-form and structured (JSON) formats. The transform under test is the commercial transform `transform` mode (top-1 surrogate, pii_entities=ALL).

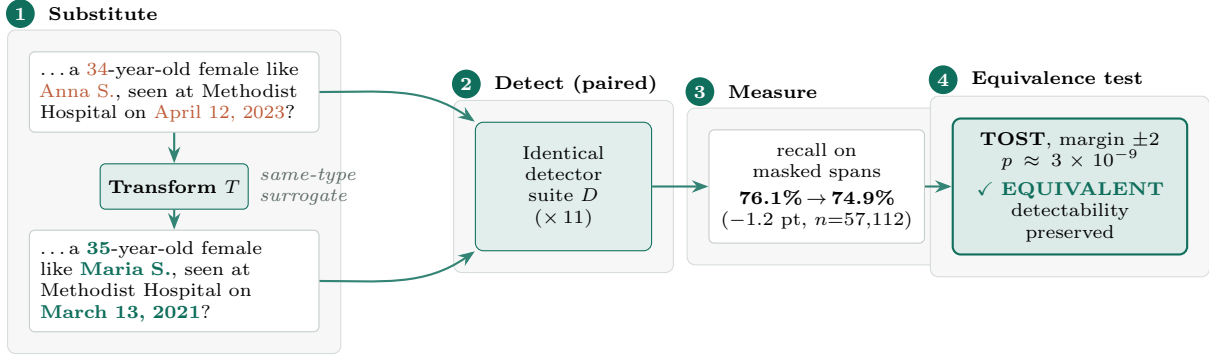


Figure 1: **The paired evaluation protocol, with a worked example.** (1) The transform replaces each PHI value with a same-type surrogate (**original** → **surrogate**), leaving everything else byte-identical. (2) The *identical* 11-detector suite scores *both* the original d and the transformed d' , so any per-span change is attributable to the substitution alone. (3) Utility is measured *only on the spans the transform masks* (overlap match), decoupling it from coverage. (4) The paired recall difference is assessed with an equivalence test (TOST, margin ± 2 pts) rather than a significance test—which the large sample would make trivially “significant” (§4). Result: the change is statistically equivalent to zero.

Comparison conditions. The core study answers “does *this* transform preserve detectability.” Three contrasts isolate *why* and *how generally*:

- **C1—Redaction upper-bound.** Replace each masked value with ***** (no surrogate signal) and re-detect; the transform–redact gap quantifies what structure preservation buys.
- **C2—Open-source surrogate baseline.** Replace the commercial generator with an open one (Faker (Faraglia and Other Contributors, 2012)) on the *same* masked spans; tests whether the result is generic to substitution or tool-specific, and makes the pipeline reproducible.
- **C3—Per-benchmark equivalence.** Run TOST within each benchmark, sweeping Δ , to check the pooled claim is not an averaging artifact.

The core paired study, the equivalence analysis (pooled and per-benchmark), the redaction floor, the open-surrogate baseline, and the error typology are complete; C1/C2 are currently demonstrated with the CPU detector (Presidio), with a full-panel extension noted as the next strengthening step.

6. Results

Detectability is statistically equivalent. Restricting to the 57,112 masked spans and pooling all 11 detectors (via the released `analyze_equivalence.py`), recall moves $76.1\% \rightarrow 74.9\%$ (-1.2 pts; 95% CI $[-1.5, -1.0]$). The TOST equivalence test ($\Delta = 2$) rejects non-equivalence at $p \approx 3 \times 10^{-9}$: the change is statistically bounded within ± 2 points of zero. Ranking is preserved; Llama 3.3-70B is essentially unchanged ($\Delta F1 +0.003$). The same contingency is “significant” under McNemar’s test—a direct demonstration of the large- N trap.

Whole-document view (conservative). Before restricting to masked spans, Table 6 reports span-level F1 and leakage on *all* gold spans, per detector. Whole-document $\Delta F1$ ranges from $+0.003$ (Llama 3.3-70B) to -0.047 (Qwen 3.5-35B-A3B), and leakage rises by only $+0.3$ to $+4.1$ points. This view is conservative by design—it mixes masked spans with exact-boundary penalties on length-changed surrogates and with spans the transform never touched—yet it already bounds the worst case (no detector’s mean F1 moves by more than 4.7 points) and preserves ranking across a $100\times$ span of detector quality ($F1\ 0.76 \rightarrow 0.04$). The masked-span analysis below isolates the substitution effect from this dilution.

Detector	Exact	Overlap
Gemma 4 31B	93.1	99.8
Qwen 3.5-9B	92.0	100.0
Qwen 3.5-35B-A3B	90.0	98.4
Llama 3.3-70B	91.4	98.3
GPT-5	91.9	98.3
Gemma 4 E4B	88.9	98.2
Qwen 3.5-4B	84.7	98.0
Presidio	91.3	97.4
DeepSeek V2-Lite	87.7	92.8

Table 1: Recall retention on masked spans (%), transformed÷original.

Recall retention on masked spans (overlap). Restricting to masked spans removes that dilution. When the transform masks a span, detectors still find the surrogate 93–100% of the time (Table 1); the ~3-point exact-boundary drop is a length-jitter artifact that vanishes under overlap matching. (The two floor detectors, Llama 3.1-8B and OBI `deid_roberta`, are omitted from Table 1: their original recall is so low that the retention ratio is dominated by noise.)

Per-benchmark equivalence (C3). Table 2 (57,112 masked spans, via `scripts/analyze_equivalence.py`) shows the pooled equivalence is *not uniform*: four benchmarks are equivalent within ± 2 points; MEDDOCAN and PII-nl require ± 3 (both dense, identifier-heavy, non-English—where surrogate generation is hardest, §7); MultiCoNER has too few masked spans (220) to test. The honest claim: *detectability is equivalent within ± 2 points on average and on clean text, and within ± 3 on the hardest identifier-dense text*—the residual is concentrated and attributable, not diffuse degradation (Figure 2).

Redaction floor (C1). Replacing masked values with ***** and re-detecting with Presidio (Table 3): transform recall tracks the *original* within a few points, while redaction collapses to ≈ 0 –4% (the residual is spurious overlap on the * run). Structure-preserving substitution preserves essentially the entire detectability that redaction destroys.

Open-surrogate baseline (C2). Replacing the commercial generator with Faker on the same masked spans (Table 4) yields two findings. *Generality*: an open substitutor preserves

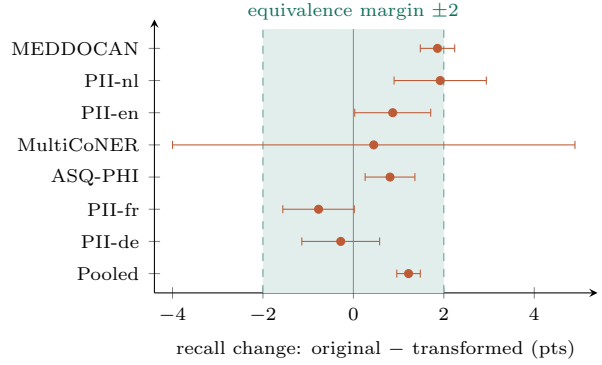


Figure 2: Per-benchmark equivalence (C3). Masked-span recall change (original – transformed) with 95% CIs; the shaded band is the ± 2 -pt equivalence margin. The pooled estimate and five of seven benchmarks sit fully inside; only MEDDOCAN and PII-nl reach past +2 (hence a ± 3 margin). MultiCoNER’s wide interval reflects its 220 masked spans.

Benchmark	Masked	O→T	McN. χ^2	Margin
ASQ-PHI	5,427	91.9→91.1	8.0	± 2 ✓
MEDDOCAN	31,978	73.4→71.5	92.4	± 3 ✓
MultiCoNER	220	71.8→71.4	0.0	underpow.
PII en	5,181	76.0→75.1	4.0	± 2 ✓
PII nl	3,757	78.2→76.3	13.1	± 3 ✓
PII fr	5,984	75.2→75.9	3.5	± 2 ✓
PII de	4,565	76.3→76.5	0.4	± 2 ✓
Pooled	57,112	76.1→74.9	82.1	± 2 ✓

Table 2: Per-benchmark equivalence. O→T = masked-span recall, original→transform (%). Pooled TOST $\Delta=2$: $p=3\times 10^{-9}$.

masked-span recall at least as well as the original—detectability preservation appears to be a property of well-formed same-type substitution, not of one vendor, and reproduces without proprietary access. *The residual is generator quality*: Faker, emitting clean canonical values, *exceeds* the commercial transform by 18–20 points on the hard non-English benchmarks, precisely where the commercial surrogates truncate or garble (§7).

Leakage barely moves (+0.3 to +4.1 pts across detectors; Table 6), so surrogates are not systematically easier to miss than the PHI they replace.

Benchmark	Masked	Orig.	Transf.	Redact
ASQ-PHI	531	94.5	95.7	0.0
MEDDOCAN	2,925	75.4	72.9	0.0
PII en	471	75.6	73.5	1.9
PII nl	373	83.6	77.7	3.8
PII fr	544	61.4	59.9	2.2
PII de	415	56.1	58.3	0.0

Table 3: C1 redaction floor: Presidio masked-span recall (%).

Benchmark	Masked	Orig.	the transform	Faker
ASQ-PHI	531	94.5	95.7	97.7
MEDDOCAN	2,925	75.4	72.9	79.7
PII en	471	75.6	73.5	89.2
PII nl	373	83.6	77.7	93.0
PII fr	544	61.4	59.9	78.5
PII de	415	56.1	58.3	78.1

Table 4: C2 open-surrogate baseline: Presidio masked-span recall (%). Detector-independent argument; Presidio-only demonstration.

Where the loss lands (per benchmark). Table 7 breaks whole-document $\Delta F1$ out by benchmark and detector. The pattern is sharp: almost all of the loss concentrates on **ASQ-PHI**—short, sparse-PHI adversarial queries where a single substitution dominates the document score (-0.09 to -0.18)—while the other six benchmarks are essentially flat ($|\Delta F1| \leq 0.06$, most ≤ 0.03). Even the fine-tuned OBI tagger and the rule-based Presidio move by at most a few thousandths on most benchmarks. This is the first sign that the effect tracks *surrogate-generation difficulty* (short adversarial text, dense non-English identifiers) rather than detector family—made precise in the error analysis (§7).

What the transform does (examples). Table 5 shows the substitution qualitatively across languages and formats: each PHI value becomes a same-type surrogate while clinical shorthand, foreign-language syntax, and JSON structure are left byte-for-byte intact. On the well-formed English cases every surrogate is still detected across most of the panel.

Coverage, reported separately. All results above are measured *on the spans the transform masks*. Coverage—how much true PHI it de-

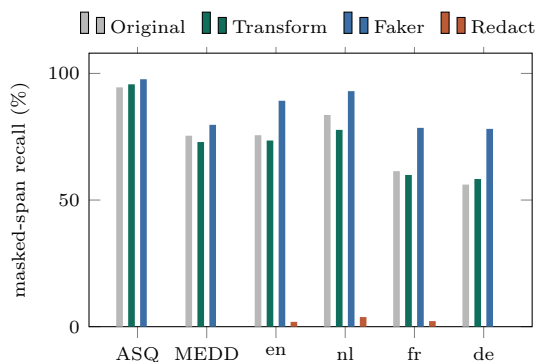


Figure 3: What structure preservation buys (C1/C2; Presidio masked-span recall). Transform tracks the original, the open Faker baseline is at least as high, and redaction collapses to ≈ 0 . The gap between the Transform/Faker bars and the Redact bar is the detectability that structure preservation retains.

tests and replaces—is a distinct, detector-side axis. It tracks how closely a benchmark’s annotation matches the transform’s notion of sensitive content: it masks **80.9%** of gold PHI on ASQ-PHI and **48.3%** on MEDDOCAN, and less ($\approx 26\%$) on general-domain NER/PII corpora whose annotated entities (encyclopedic names, generic places) fall outside that scope; a configuration sweep confirmed **domain=General** maximizes coverage. A per-entity-type breakdown and a detection-vs-replacement diagnosis (the coverage gap is mostly a *replacement*-step issue— $\approx 77\%$ of missed clinical identifiers were flagged by the transform’s own detector but not substituted) are in Appendix B. Decoupling matters because the two failure modes have different owners and fixes: an under-masked span is a *detection/replacement* miss, whereas a masked-but-missed surrogate is a *generation* defect (§7). Conflating them on whole-document F1 (Table 6) would let a high-coverage/low-quality transform and a low-coverage/high-quality one look identical. We therefore report coverage as context and reserve the utility claim for masked spans.

Matching modes. Unless noted, retention uses *overlap* matching (detector flags any part of the surrogate span). Exact matching (Table 1, left) additionally requires identical character

Case	Original	Transformed
ASQ-PHI (clinical query)	...a 34-year-old female ...like Anna S., ...at Methodist Hospital on April 12, 2023?	...a 35-year-old female ...like Maria S., ...at Methodist Hospital on March 13, 2021?
ASQ-PHI (short-hand)	Rec mgmt of 70yo M w/ CHF, seen by Dr. John L. at Mt. Sinai on Feb 21, 2023. alt tx options...	Rec mgmt of 73yo M w/ CHF, seen by Dr. James L. at Mt. Egypt on Nov 19, 2021. alt tx options...
PII (German)	...ermächtigen hiermit Monsignore, ...Mit Datum 23/07/2011...	...ermächtigen hiermit Fulgenzio, ...Mit Datum 24/07/2011...
PII (French, JSON)	{"Date": "20/05/2022", "City": "Saint-Priest", "Username": "phprosdocimo"}	{"Date": "21/05/2023", "City": "Saint-Priest", "Username": "phprosdocimo"}

Table 5: Worked transform examples. Only the sensitive value moves; abbreviations (w/ CHF, alt tx), non-English syntax, and JSON keys/quotes/indentation are preserved, so downstream parsers and detectors keep working. In the JSON case the transform changes only the date value and leaves the (also-sensitive-looking) username untouched—a coverage decision, not a utility one (see below).

boundaries and is therefore sensitive to surrogate length changes (“Anna S.”→“Maria S.” shifts the end offset); the ~3-point exact–overlap gap is this boundary jitter, not missed PHI. Type matching (boundary + type) sits between the two and tracks overlap closely.

7. Error Analysis

The lost population. Pooled across detectors and masked spans: 40,165 (span×detector) pairs found in both conditions; 3,300 lost. Half (50%) of lost spans are the *same length* as the original—so this is not a boundary effect. By type: LOCATION 27%, NAME 23%, DATE/AGE 22%, ID/contact 19%.

Three failure modes—all generator-side. (1) *Malformed/truncated surrogates* (largest cause): Chicago→Illino, El Paso→El, Ciudad de la Habana→Cuidad de la Havana. The fragment no longer matches the lexical pattern detectors learned for real names/places; this is why span-level loss is highest on MEDDOCAN (6.9%; Spanish, dense, identifier-heavy) and lowest on clean English ASQ-PHI (2.5%). (2) *Loss of salience*: a canonical entity replaced by an obscure one (Cedars-Sinai→Vidant); detectors partly rely on pre-training familiarity, so swapping a famous value for a rare one removes the prior. Inherent to any value substitution; mainly costs

weaker detectors. (3) *x-masking of IDs/emails*: nachorutor@...→nxxxxxxxxx@.... The x run preserves format but breaks the realistic-token pattern. This case is *privacy-positive*—the original value is destroyed—even though it counts against recall.

Implication. Detectors are not getting worse at PHI; the small recall gap is driven by surrogate-generation quality (truncation, garbling, salience, x-masking)—generating well-formed, in-distribution synthetic text is itself a hard problem in text data augmentation (Bao et al., 2024b). C2 confirms this directly: an open generator emitting clean canonical values *exceeds* the commercial transform by 18–20 points on exactly the hard non-English benchmarks where its surrogates truncate—so the residual tracks generator quality, not the act of substitution.

8. Discussion

What the evidence supports. Structure-preserving substitution does not hide well-formed PHI from downstream detection. Because the effect is equivalence-bounded within ± 2 points and ranking-preserving across a heterogeneous 11-detector panel, a transform of this quality can be inserted ahead of detection/analytics pipelines built for real clinical text without materially degrading downstream detection. **Generality (C2).** Faker preserves masked-span recall at least

as well as the original across all six benchmarks, so ± 2 points appears to be a property of well-formed substitution rather than of one vendor; where the commercial generator trails, a cleaner generator closes the gap, locating the residual squarely in generation quality. **A reusable protocol.** The paired masked-span design and the TOST margin are not specific to one vendor or language—a template for auditing any structure-preserving privacy transform.

9. Ethics and Data Statement

All benchmarks are public or synthetic (ASQ-PHI synthetic; MEDDOCAN released for a shared task; PII-Masking-300k synthetic; MultiCoNER v2 public). **No real patient data is used.** We release the 250-document subsets and scoring code with license notes. **Dual-use.** A utility-preserving de-identifier could in principle launder identifiable data into a fluent form; our masked-span/leakage reporting and the privacy-positive framing of x-masking keep the privacy accounting explicit. **Conflict of interest.** The commercial transform evaluated as the case study is developed by a vendor with which one or more authors are affiliated (details withheld for anonymous review); the work was conducted with that vendor’s support. To limit bias we (i) frame the contribution as a reusable protocol, (ii) include open-source baselines so results are reproducible without proprietary access, and (iii) report leakage and coverage alongside utility.

10. Limitations

Coverage is reported but not the focus; a transform can preserve utility on what it masks while under-masking (the axes are independent by design). 250 docs/benchmark bounds per-benchmark power (though the pooled masked-span N is large). C1/C2 are currently demonstrated with a CPU detector (Presidio); a full 11-detector extension is future work. LLM detectors are prompt-sensitive and their behaviour shifts with task framing and structural variation (Gendron et al., 2024; Bao et al., 2024a); we fix one prompt per model and release it with the code and data at the project’s reproducibility page. Finally, all seven benchmarks are public or synthetic; validating the protocol on real-EHR

Detector	F1		Δ		Leak	
	orig	transf	F1	orig	transf	
(lr)2-3(lr)5-6						
Gemma 4 31B	.755	.710	−.045	.255	.285	
Gemma 4 E4B	.737	.698	−.038	.276	.311	
Llama 3.3-70B	.725	.728	+ .003	.245	.262	
Qwen 3.5-35B-A3B	.715	.668	−.047	.254	.295	
OpenAI GPT-5	.705	.674	−.032	.308	.333	
Qwen 3.5-9B	.655	.621	−.034	.391	.422	
Qwen 3.5-4B	.567	.533	−.034	.483	.515	
Presidio	.416	.398	−.018	.553	.570	
DeepSeek V2-Lite	.409	.384	−.025	.672	.696	
Llama 3.1-8B	.391	.376	−.015	.633	.650	
OBI <code>deid_roberta</code>	.041	.040	−.002	.941	.944	

Table 6: Whole-document view (all 11 detectors), mean over 7 benchmarks. Span-level F1 (type match) and leakage (1−recall). Conservative: it includes untouched spans and boundary penalties. Ranking is preserved and $\Delta F1$ is bounded.

corpora (n2c2, MIMIC-IV-Note, CARMEN-I) is future work pending the relevant data-use agreements.

Appendix A. Additional Results

Appendix B. Coverage Details

Coverage (§6) is the fraction of a benchmark’s gold PHI whose characters the transform actually changed (difflib alignment). Table 8 breaks it down by entity type on the clinical sets (ASQ-PHI + MEDDOCAN). High-frequency free-text types (names, dates) are masked well; the weak spots are high-sensitivity structured identifiers and geography—exactly the HIPAA Safe Harbor items that most need masking.

Detection vs. replacement. On a sample of 8 MEDDOCAN documents (62 missed ID/location spans), 48/62 ($\approx 77\%$) of the *unmasked* identifiers were nonetheless flagged as sensitive by the transform’s own detector—they were detected but not substituted; only 14/62 (23%) were undetected. The coverage gap is therefore mostly a *replacement-step* issue (act on everything the detector surfaces), which is more tractable than raising detection recall. (Small sample; “detected” judged by loose token overlap, so 77% is directional.) Concrete unmasked HIPAA identifiers included patient IDs (80926,

Detector	ASQ-PHI	MEDDOCAN	PII en	PII nl	PII fr	PII de	MultiCoNER
Gemma 4 31B	-.175	-.017	-.043	-.038	-.014	-.016	-.009
Gemma 4 E4B	-.106	-.060	-.032	-.021	-.018	-.025	-.007
Qwen 3.5-35B-A3B	-.132	-.034	-.026	-.034	-.040	-.054	-.007
OpenAI GPT-5	-.098	-.046	-.049	-.019	+.008	-.005	-.014
Qwen 3.5-9B	-.123	-.017	-.025	-.010	-.034	-.024	-.008
Qwen 3.5-4B	-.086	-.055	-.034	-.028	-.015	-.023	+.002
Presidio	+.003	-.037	-.032	-.015	-.020	-.015	-.007
Llama 3.1-8B	-.108	+.023	-.014	+.007	-.009	+.003	-.006
OBI deid_roberta	+.001	-.008	-.004	-.001	+.003	-.002	-.001

Table 7: Per-benchmark whole-document $\Delta F1$ (transformed – original; negative = drop). Almost all loss lands on ASQ-PHI; the other six benchmarks are flat. Llama 3.3-70B and DeepSeek V2-Lite omitted for space (both flat; see Table 6 for their pooled $\Delta F1$).

Entity type (clinical)	Coverage	over-fragments boundaries and false-fires on the
NAME / person	75.0%	age 60. The surrogate values shown are the trans-
DATE / age	73.4%	form’s output, not the original PHI.
ID / contact (MRN, patient ID, phone, email)	50.7%	
ORG / facility	46	Gold (reference) 4 PHI spans
LOCATION / address	43	Evaluation of long-term outcomes for bypass surgery

Table 8: Coverage by entity type on clinical benchmarks. Overall the transform altered 39% of annotated PII (54% on clinical sets); general-domain corpora sit near 26%.

7845693), care-contact IDs (4387684), and facilities/geography (Hospital de Cruces, España, postal codes 41005, 28047).

Appendix C. Representative Per-Detector Examples

All-detector overlay on one document (“ranking proof”). All eleven detectors’ predictions on one transformed document (asq_00001), overlaid on the text and sorted by this document’s F1: **green** = correctly caught PHI, **yellow** = false positive, **red** = missed PHI (leakage). The four gold surrogate values are John T., the *x*-masked location Saint. Vxxxxxxx, April 25th, 2018, and the ID 987654321. Eight of eleven detectors tag all four cleanly (F1 100); the two floor detectors, Presidio and DeepSeek, miss the *x*-masked location surrogate (§7 mode 3)—DeepSeek also drops the numeric ID—and OBI

Gold (reference)	4 PHI spans
Evaluation of long-term outcomes for bypass surgery in patients over 60, referencing Mr. John T., operated at Saint. Vxxxxxxx on April 25th, 2018 (ID: 987654321)?	
OpenAI GPT-5	tp 4 · fp 0 · missed 0 · F1 100%
Evaluation of long-term outcomes for bypass surgery in patients over 60, referencing Mr. John T., operated at Saint. Vxxxxxxx on April 25th, 2018 (ID: 987654321)?	
Llama 3.1-8B	tp 4 · fp 0 · missed 0 · F1 100%
Evaluation of long-term outcomes for bypass surgery in patients over 60, referencing Mr. John T., operated at Saint. Vxxxxxxx on April 25th, 2018 (ID: 987654321)?	
Qwen 3.5-4B	tp 4 · fp 0 · missed 0 · F1 100%
Evaluation of long-term outcomes for bypass surgery in patients over 60, referencing Mr. John T., operated at Saint. Vxxxxxxx on April 25th, 2018 (ID: 987654321)?	
Gemma 4 31B	tp 4 · fp 0 · missed 0 · F1 100%
Evaluation of long-term outcomes for bypass surgery in patients over 60, referencing Mr. John T., operated at Saint. Vxxxxxxx on April 25th, 2018 (ID: 987654321)?	
Llama 3.3-70B	tp 4 · fp 0 · missed 0 · F1 100%
Evaluation of long-term outcomes for bypass surgery in patients over 60, referencing Mr. John T., operated at Saint. Vxxxxxxx on April 25th, 2018 (ID: 987654321)?	
Qwen 3.5-35B-A3B	tp 4 · fp 0 · missed 0 · F1 100%
Evaluation of long-term outcomes for bypass surgery in patients over 60, referencing Mr. John T., operated at Saint. Vxxxxxxx on April 25th, 2018 (ID: 987654321)?	
Gemma 4 E4B	tp 4 · fp 0 · missed 0 · F1 100%
Evaluation of long-term outcomes for bypass surgery in patients over 60, referencing Mr. John T., operated at Saint. Vxxxxxxx on April 25th, 2018 (ID: 987654321)?	

Qwen 3.5-9B tp 4 · fp 0 · missed 0 · F1 100%
 Evaluation of long-term outcomes for bypass surgery in patients over 60, referencing Mr. John T., operated at Saint. Vxxxxxxx on April 25th, 2018 (ID: 987654321)?

OBI deid.roberta tp 4 · fp 1 · missed 0 · F1 89%
 Evaluation of long-term outcomes for bypass surgery in patients over 60, referencing Mr. John T., operated at Saint. Vxxxxxxx on April 25th, 2018 (ID: 987654321)?

Microsoft Presidio tp 3 · fp 1 · missed 1 · F1 75%
 Evaluation of long-term outcomes for bypass surgery in patients over 60, referencing Mr. John T., operated at Saint. Vxxxxxxx on April 25th, 2018 (ID: 987654321)?

DeepSeek V2-Lite tp 2 · fp 0 · missed 2 · F1 67%
 Evaluation of long-term outcomes for bypass surgery in patients over 60, referencing Mr. John T., operated at Saint. Vxxxxxxx on April 25th, 2018 (ID: 987654321)?

A harder case: a dense Spanish clinical header. Figure-style overlay on a 260-character window of a MEDDOCAN document (S0210-...008-1), where the same eleven detectors diverge far more—F1 from 96 down to 0. On dense, identifier-heavy non-English text the *per-document* ranking reshuffles (here DeepSeek and Qwen-4B lead, while Llama 3.1-8B finds nothing in this window), even though the *aggregate* ranking over 250 documents is stable (Table 6). The garbled surrogate Ciudad Real (from Ciudad Real) is still caught by most detectors; the misses cluster on the professional-licence ID 03 14 16485, the sex fields, and the repeated age—the identifier types with the weakest coverage and the hardest surrogates (Appendix B).

Gold (reference) 12 PHI spans
 Ciudad Real. CP: 13002. Datos asistenciales.
 Referencia de nacimiento: 24/10/1963. País:
 España. Mejores: 53 años Sexo: H. Referencia de
 Ingreso: 14/12/2017. Episodio: 746589123. Médico:
 Luis Ruiz Camuñas NºCol: 03 14 16485. Historia
 Actual: Varón de 53 años q

DeepSeek V2-Lite tp 11 · fp 0 · missed 1 · F1 96%
 Ciudad Real. CP: 13002. Datos asistenciales.
 Referencia de nacimiento: 24/10/1963. País:
 España. Mejores: 53 años Sexo: H. Referencia de
 Ingreso: 14/12/2017. Episodio: 746589123. Médico:
 Luis Ruiz Camuñas NºCol: 03 14 16485. Historia
 Actual: Varón de 53 años q

Qwen 3.5-4B tp 11 · fp 0 · missed 1 · F1 96%
 Ciudad Real. CP: 13002. Datos asistenciales.
 Referencia de nacimiento: 24/10/1963. País:
 España. Mejores: 53 años Sexo: H. Referencia de
 Ingreso: 14/12/2017. Episodio: 746589123. Médico:
 Luis Ruiz Camuñas NºCol: 03 14 16485. Historia
 Actual: Varón de 53 años q

OBI deid.roberta tp 10 · fp 0 · missed 2 · F1 91%
 Ciudad Real. CP: 13002. Datos asistenciales.
 Referencia de nacimiento: 24/10/1963. País:
 España. Mejores: 53 años Sexo: H. Referencia de
 Ingreso: 14/12/2017. Episodio: 746589123. Médico:
 Luis Ruiz Camuñas NºCol: 03 14 16485. Historia
 Actual: Varón de 53 años q

Qwen 3.5-9B tp 10 · fp 0 · missed 2 · F1 91%
 Ciudad Real. CP: 13002. Datos asistenciales.
 Referencia de nacimiento: 24/10/1963. País:
 España. Mejores: 53 años Sexo: H. Referencia de
 Ingreso: 14/12/2017. Episodio: 746589123. Médico:
 Luis Ruiz Camuñas NºCol: 03 14 16485. Historia
 Actual: Varón de 53 años q

Gemma 4 31B tp 9 · fp 0 · missed 3 · F1 86%
 Ciudad Real. CP: 13002. Datos asistenciales.
 Referencia de nacimiento: 24/10/1963. País:
 España. Mejores: 53 años Sexo: H. Referencia de
 Ingreso: 14/12/2017. Episodio: 746589123. Médico:
 Luis Ruiz Camuñas NºCol: 03 14 16485. Historia
 Actual: Varón de 53 años q

Qwen 3.5-35B-A3B tp 9 · fp 1 · missed 3 · F1 82%
 Ciudad Real. CP: 13002. Datos asistenciales.
 Referencia de nacimiento: 24/10/1963. País:
 España. Mejores: 53 años Sexo: H. Referencia de
 Ingreso: 14/12/2017. Episodio: 746589123. Médico:
 Luis Ruiz Camuñas NºCol: 03 14 16485. Historia
 Actual: Varón de 53 años q

OpenAI GPT-5 tp 8 · fp 0 · missed 4 · F1 80%
 Ciudad Real. CP: 13002. Datos asistenciales.
 Referencia de nacimiento: 24/10/1963. País:
 España. Mejores: 53 años Sexo: H. Referencia de
 Ingreso: 14/12/2017. Episodio: 746589123. Médico:
 Luis Ruiz Camuñas NºCol: 03 14 16485. Historia
 Actual: Varón de 53 años q

Llama 3.3-70B tp 8 · fp 0 · missed 4 · F1 80%
 Ciudad Real. CP: 13002. Datos asistenciales.
 Referencia de nacimiento: 24/10/1963. País:
 España. Mejores: 53 años Sexo: H. Referencia de
 Ingreso: 14/12/2017. Episodio: 746589123. Médico:
 Luis Ruiz Camuñas NºCol: 03 14 16485. Historia
 Actual: Varón de 53 años q

Gemma 4 E4B tp 8 · fp 0 · missed 4 · F1 80%
 Ciudad Real. CP: 13002. Datos asistenciales.
 Referencia de nacimiento: 24/10/1963. País:
 España. Mejores: 53 años Sexo: H. Referencia de
 Ingreso: 14/12/2017. Episodio: 746589123. Médico:
 Luis Ruiz Camuñas NºCol: 03 14 16485. Historia
 Actual: Varón de 53 años q

Microsoft Presidio tp 6 · fp 1 · missed 6 · F1 63%
 Ciudad Real. CP: 13002. Datos asistenciales.
 Referencia de nacimiento: 24/10/1963. País:
 España. Mejores: 53 años Sexo: H. Referencia de
 Ingreso: 14/12/2017. Episodio: 746589123. Médico:
 Luis Ruiz Camuñas NºCol: 03 14 16485. Historia
 Actual: Varón de 53 años q

Llama 3.1-8B tp 0 · fp 0 · missed 12 · F1 0%
 Ciudad Real. CP: 13002. Datos asistenciales.
 Referencia de nacimiento: 24/10/1963. País:
 España. Mejores: 53 años Sexo: H. Referencia de
 Ingreso: 14/12/2017. Episodio: 746589123. Médico:
 Luis Ruiz Camuñas NºCol: 03 14 16485. Historia
 Actual: Varón de 53 años q

Agreement on well-formed surrogates.
 On clean English (ASQ-PHI), a plausible same-

type surrogate is caught by nearly the whole panel (Table 9): the substitution is invisible to detection. Divergence is confined to the two floor detectors (Llama 3.1-8B, DeepSeek V2-Lite).

Document	Surrogate span (type)	Caught
asq-00000	Maria S. (NAME)	10/11
asq-00000	Methodist Hospital (LOC)	9/11
asq-00000	March 13, 2021 (DATE)	9/11
asq-00003	James L. (NAME)	11/11
asq-00003	Mt. Egypt (LOC)	11/11
asq-00003	Nov 19, 2021 (DATE)	11/11

Table 9: Number of detectors (of 11) that find each well-formed surrogate. Same-type swaps stay broadly detectable.

Representative losses, by failure type. Table 10 shows genuine “lost” cases (MEDDOCAN): the original value was found by most detectors, but a defective surrogate is found by few. Each maps to one of the three failure types in §7, and the drop is shared across the panel—i.e. it is a property of the surrogate, not of any one detector.

Original → surrogate	O → T	Type
Cuba → Havana	9 → 2	salience
San Fernando → Francsico Luis	8 → 3	garbled
Cádiz → Cadiz	8 → 4	accent stripped
Cecilio Pujazón → Pujazón	8 → 4	truncation
ignaciotorne@... → ixxxxxxxxxxx@...	9 → 2	x-mask
23/07/1948 → 2xxxxxxxx	11 → 4	x-mask
40 → 35 (age)	9 → 2	bare number

Table 10: Representative lost spans on MEDDOCAN. O → T = detectors finding the original vs. the surrogate (panel of 11–12; original includes one extra decoding variant). Losses are shared across detectors and align with the §7 typology.

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